100 Scientific Terms and Definitions

A collection of 100 scientific terms with detailed, expanded definitions for study and revision.

1. Atom

An atom is the basic unit of an element and is the smallest particle that still retains the chemical identity of that element. It contains a dense nucleus made of protons and neutrons, surrounded by electrons. The number of protons determines which element the atom represents. Atoms can join with other atoms through chemical bonds to form molecules and compounds.

2. Molecule

A molecule is a group of two or more atoms that are chemically bonded together and behave as a single unit. The atoms may be from the same element, such as oxygen molecules, or from different elements, such as water. The arrangement and bonding of atoms determine the molecule's physical and chemical properties. Molecules are fundamental units of many substances found in nature.

3. Element

An element is a pure chemical substance consisting of atoms that all have the same number of protons in their nuclei. Each element has a unique atomic number and chemical symbol. Elements cannot be broken into simpler substances by ordinary chemical reactions. The periodic table organizes known elements according to their atomic properties.

4. Compound

A compound is a pure substance formed when atoms of two or more different elements combine chemically in fixed proportions. The resulting substance usually has properties different from those of the elements that formed it. Compounds can be separated into their constituent elements through suitable chemical processes. Water, carbon dioxide, and sodium chloride are common examples.

5. Matter

Matter is anything that has mass and occupies physical space. It exists in different states, including solids, liquids, gases, and under special conditions plasma. Matter is made of particles such as atoms and molecules that are constantly in motion. Changes in temperature, pressure, and energy can cause matter to change from one state to another.

6. Mass

Mass is a physical quantity that represents the amount of matter in an object. It is commonly measured in kilograms in the International System of Units. Mass is different from weight because weight depends on gravitational force, while mass remains essentially constant. Mass also affects an object's resistance to changes in its motion.

7. Force

Force is a push or pull that can change the motion, direction, or shape of an object. Forces may result from direct contact, such as friction, or act at a distance, such as gravity and magnetism. Force is measured in newtons. According to Newton's laws, the net force acting on an object determines how its motion changes.

8. Energy

Energy is the capacity of a physical system to perform work or bring about a change. It can exist in many forms, including kinetic, potential, thermal, chemical, electrical, and nuclear energy. Energy can be transferred or transformed from one form to another. The law of conservation of energy states that energy cannot be created or destroyed in an isolated system.

9. Work

In physics, work is the transfer of energy that occurs when a force causes an object to move through a displacement. The amount of work depends on the force, displacement, and angle between them. Work is measured in joules. If a force acts but produces no displacement in the direction of the force, no mechanical work is done.

10. Power

Power is the rate at which work is performed or energy is transferred. It describes how quickly energy is used or delivered rather than the total amount of energy involved. Power is measured in watts, with one watt equal to one joule per second. Electrical appliances, engines, and machines can all be compared using their power ratings.

11. Velocity

Velocity describes how quickly an object changes its position and includes the direction of motion. It is therefore a vector quantity, unlike speed, which only gives the magnitude of motion. Velocity can change when an object speeds up, slows down, or changes direction. It is commonly measured in metres per second.

12. Acceleration

Acceleration is the rate at which an object's velocity changes with time. An object accelerates whenever its speed or direction changes. Acceleration is a vector quantity and can occur even when an object's speed remains constant if its direction changes. Its SI unit is metres per second squared.

13. Momentum

Momentum is a measure of the quantity of motion possessed by a moving object. It depends on both the object's mass and its velocity and is calculated as mass multiplied by velocity. Momentum is a vector quantity because it has direction. In an isolated system, total momentum remains conserved during interactions.

14. Gravity

Gravity is the attractive interaction between objects that have mass. On Earth, gravity gives objects weight and causes unsupported objects to accelerate toward the ground. Gravity also keeps planets in orbit around stars and moons in orbit around planets. Its strength depends on the masses involved and the distance between them.

15. Pressure

Pressure describes how much force is applied over a given area of a surface. It is calculated by dividing perpendicular force by area and is measured in pascals. Pressure is important in gases, liquids, weather systems, machines, and biological processes. The same force produces greater pressure when it acts over a smaller area.

16. Density

Density is a measure of how much mass is contained in a given volume of a substance. It is calculated by dividing mass by volume and is commonly expressed in kilograms per cubic metre. Density can help identify materials and predict whether objects will float or sink in a fluid. Temperature and pressure can affect the density of some substances.

17. Temperature

Temperature is a physical quantity that indicates the thermal state of a substance and is closely related to the average kinetic energy of its particles. It determines the direction in which heat naturally flows between objects. Temperature is commonly measured in degrees Celsius, kelvin, or Fahrenheit. It influences many physical and chemical processes.

18. Heat

Heat is energy transferred from one body or system to another because of a temperature difference. It naturally flows from a region of higher temperature toward a region of lower temperature. Heat can be transferred by conduction, convection, or radiation. Unlike temperature, heat refers to energy in transfer rather than a measure of thermal state.

19. Nucleus

The nucleus is the extremely dense central part of an atom. It contains positively charged protons and electrically neutral neutrons, which together account for almost all of an atom's mass. The number of protons identifies the element. Nuclear reactions can release very large amounts of energy by changing the nucleus.

20. Electron

An electron is a subatomic particle with a negative electric charge. Electrons occupy regions around an atom's nucleus and are responsible for many chemical and electrical properties of matter. Their arrangement determines how atoms interact and form chemical bonds. The movement of electrons through conductors produces electric current.

21. Proton

A proton is a positively charged subatomic particle located inside the nucleus of an atom. The number of protons in an atom is called its atomic number and determines the identity of the element. Protons are composed of smaller particles called quarks. Their positive charge is equal in magnitude to the negative charge of an electron.

22. Neutron

A neutron is a subatomic particle found in the nucleus of most atoms and has no net electric charge. It contributes significantly to the mass of an atom and helps influence nuclear stability. Atoms of the same element can contain different numbers of neutrons, forming isotopes. Free neutrons can also participate in nuclear reactions.

23. Ion

An ion is an atom or molecule that carries an electrical charge because it has gained or lost one or more electrons. Losing electrons produces a positively charged ion called a cation, while gaining electrons produces a negatively charged anion. Ions are important in chemical reactions, electrical conduction, and biological processes.

24. Isotope

An isotope is a form of an element that has the same number of protons but a different number of neutrons. Because isotopes have the same atomic number, they belong to the same element, but their masses differ. Some isotopes are stable, while others are radioactive. Isotopes are used in medicine, scientific research, and dating ancient materials.

25. Chemical Reaction

A chemical reaction is a process in which atoms are rearranged to transform one or more starting substances into new substances. Chemical bonds may be broken, formed, or rearranged during the process. Reactions can release or absorb energy. Examples include combustion, rusting, digestion, and the reactions involved in photosynthesis.

26. Catalyst

A catalyst is a substance that increases the rate of a chemical reaction without being permanently consumed by the reaction. It works by providing an alternative reaction pathway that generally requires less activation energy. Catalysts are important in industrial chemistry, environmental processes, and biological systems. Enzymes are catalysts that perform this role in living organisms.

27. Acid

An acid is a substance that can donate hydrogen ions in a chemical reaction or increase the concentration of hydrogen ions in aqueous solution. Acids generally have a pH below 7. Strong acids can react vigorously with certain metals and bases. Examples include hydrochloric acid, sulfuric acid, and the acids naturally present in citrus fruits.

28. Base

A base is a substance that can accept hydrogen ions or produce hydroxide ions in an aqueous solution. Bases generally have a pH above 7 and can neutralize acids to form salts and water. Some bases are strong and highly reactive, while others are weak. Sodium hydroxide and ammonia are common examples of basic substances.

29. pH

pH is a numerical scale used to describe the acidity or basicity of an aqueous solution. The scale is commonly represented from 0 to 14, with lower values indicating greater acidity and higher values indicating greater basicity. A pH of about 7 is considered neutral at standard conditions. pH is important in chemistry, biology, agriculture, and environmental science.

30. Oxidation

Oxidation is a chemical process traditionally described as the loss of electrons by an atom, ion, or molecule. It can also involve an increase in oxidation state or, in some reactions, the addition of oxygen. Oxidation commonly occurs together with reduction in redox reactions. Rusting of iron is a familiar example of an oxidation process.

31. Reduction

Reduction is a chemical process involving the gain of electrons by an atom, ion, or molecule. It can also involve a decrease in oxidation state or removal of oxygen in some reactions. Reduction always occurs together with oxidation in a redox reaction. The transfer of electrons between substances is central to many chemical and biological energy processes.

32. Cell

A cell is the basic structural and functional unit of life. Some organisms consist of a single cell, while complex organisms contain many specialized cells. Cells carry out processes such as obtaining energy, making proteins, responding to signals, and reproducing. Cells may be broadly classified as prokaryotic or eukaryotic.

33. Tissue

A tissue is a group of similar or related cells organized to perform a particular function. In multicellular organisms, tissues provide structure and allow specialized activities to occur efficiently. Animal tissues include epithelial, connective, muscle, and nervous tissues. Plants also contain specialized tissues such as vascular and protective tissues.

34. Organ

An organ is a specialized structure composed of different tissues that work together to perform one or more major functions. In animals, organs include the heart, lungs, liver, and kidneys. In plants, roots, stems, and leaves are examples of organs. Organs operate as parts of larger organ systems.

35. Organism

An organism is an individual living system capable of performing essential life processes. These processes include obtaining energy, responding to the environment, maintaining internal conditions, and reproducing. Organisms range from microscopic bacteria to large plants and animals. They may be single-celled or multicellular.

36. DNA

DNA, or deoxyribonucleic acid, is the molecule that stores hereditary information in living organisms and many viruses. Its sequence of nucleotides contains instructions used to build functional biological molecules and regulate cellular activities. DNA is organized into genes and chromosomes. Accurate copying of DNA allows genetic information to be passed from one generation to another.

37. RNA

RNA, or ribonucleic acid, is a nucleic acid involved in storing, transferring, and using genetic information. Different types of RNA perform different functions, including carrying instructions from DNA and helping assemble proteins. RNA usually consists of a single strand of nucleotides. It also plays important roles in regulation and gene expression.

38. Gene

A gene is a segment of genetic material that contains information needed to produce a functional biological product or influence a trait. Genes are located on chromosomes and can be passed from parents to offspring. Variations in genes contribute to differences among individuals. The activity of genes is regulated by complex cellular mechanisms.

39. Chromosome

A chromosome is a structure made primarily of DNA and associated proteins that carries genetic information. In eukaryotic cells, chromosomes are located in the nucleus. They help organize and package long DNA molecules so that genetic material can be accurately copied and distributed during cell division. Humans normally have 46 chromosomes in most body cells.

40. Mutation

A mutation is a change in the nucleotide sequence of DNA. Mutations can arise naturally from errors during DNA replication or from environmental factors such as radiation and certain chemicals. Some mutations have little effect, while others can alter biological traits or contribute to disease. Mutations are also an important source of genetic variation.

41. Evolution

Evolution is the change in inherited characteristics of populations over generations. It occurs through processes such as mutation, natural selection, genetic drift, and gene flow. Evolution can produce adaptations and contribute to the formation of new species. Modern evolutionary biology combines genetics with observations of populations and fossils.

42. Natural Selection

Natural selection is a mechanism of evolution in which individuals with heritable characteristics that improve survival or reproduction tend to leave more offspring. Over many generations, advantageous traits can therefore become more common in a population. The process depends on variation, inheritance, and differences in reproductive success. Natural selection does not require organisms to consciously adapt.

43. Photosynthesis

Photosynthesis is a biological process used by plants, algae, and some microorganisms to convert light energy into chemical energy. During the process, carbon dioxide and water are used to produce energy-rich organic molecules, with oxygen commonly released as a by-product. Photosynthesis occurs mainly in chloroplasts of plant cells. It is fundamental to most food chains on Earth.

44. Respiration

Cellular respiration is a series of biochemical reactions through which cells release usable energy from nutrients such as glucose. In aerobic respiration, oxygen is commonly used and carbon dioxide and water are produced. The released energy is captured largely in molecules of ATP. Respiration provides energy needed for movement, growth, repair, and other cellular activities.

45. Enzyme

An enzyme is a biological catalyst that accelerates a specific chemical reaction without being consumed. Most enzymes are proteins, although some RNA molecules can also act as catalysts. Enzymes bind particular substrates at active sites and help reactions occur efficiently under cellular conditions. Temperature and pH can strongly influence enzyme activity.

46. Protein

A protein is a large biological molecule made from chains of amino acids. The sequence of amino acids determines how a protein folds and functions. Proteins perform many roles, including catalyzing reactions, transporting substances, providing structure, and transmitting signals. Examples include enzymes, antibodies, collagen, and hemoglobin.

47. Carbohydrate

A carbohydrate is an organic compound made primarily from carbon, hydrogen, and oxygen. Carbohydrates include simple sugars such as glucose as well as complex molecules such as starch and cellulose. They are important sources of energy and can also serve structural roles. Plants produce many carbohydrates during photosynthesis.

48. Lipid

A lipid is a broad group of organic molecules that are generally insoluble in water. Lipids include fats, oils, phospholipids, and steroids. They are important for long-term energy storage, cell membrane structure, insulation, and signaling. Their chemical properties depend on the types and arrangements of their constituent molecules.

49. Vitamin

A vitamin is an organic compound required by an organism in relatively small amounts for normal growth, metabolism, and health. Humans generally obtain vitamins from food, although some can be produced or modified within the body. Different vitamins have different functions and deficiency effects. Vitamins are commonly classified as water-soluble or fat-soluble.

50. Hormone

A hormone is a chemical messenger produced by specialized cells or glands that travels to target cells and influences their activity. Hormones help regulate processes such as growth, metabolism, reproduction, stress responses, and development. They may travel through the bloodstream or act locally. Examples include insulin, adrenaline, and thyroid hormones.

51. Neuron

A neuron is a specialized cell of the nervous system that receives, processes, and transmits information. Neurons communicate using electrical signals and chemical messengers called neurotransmitters. Most neurons have structures such as dendrites for receiving signals and axons for transmitting them. Networks of neurons allow organisms to sense and respond to their environment.

52. Ecosystem

An ecosystem is a functional system made up of living organisms and the nonliving physical environment with which they interact. It includes producers, consumers, decomposers, water, soil, air, light, and nutrients. Energy generally flows through an ecosystem while materials are recycled. Ecosystems can range from small ponds to large forests and oceans.

53. Biodiversity

Biodiversity refers to the variety of life at genetic, species, and ecosystem levels. High biodiversity can contribute to ecosystem stability and provides resources such as food, medicines, and genetic material. Biodiversity can be affected by habitat loss, pollution, climate change, invasive species, and overexploitation. Conservation efforts aim to protect biological diversity.

54. Habitat

A habitat is the natural environment in which an organism or population lives and obtains the resources needed for survival. It provides suitable conditions such as food, water, shelter, temperature, and space. Different species are adapted to different habitats. Changes to habitat conditions can strongly affect population size and survival.

55. Population

In biology, a population is a group of individuals of the same species living in a particular area and potentially interacting or reproducing with one another. Population size and structure can change because of births, deaths, immigration, and emigration. Scientists study populations to understand growth, distribution, genetic variation, and ecological relationships.

56. Community

An ecological community consists of populations of different species living and interacting within the same area. Species in a community may compete, cooperate, prey on one another, or have other relationships. The composition of a community changes over time in response to environmental conditions. Communities form one biological component of ecosystems.

57. Food Chain

A food chain is a sequence that represents the transfer of energy and nutrients from one organism to another through feeding. It usually begins with a producer and continues through different levels of consumers. Energy is lost at each transfer, often as heat. Food chains are connected to form more complex food webs.

58. Food Web

A food web is a network of interconnected food chains that describes feeding relationships within an ecosystem. It shows that organisms often eat several different types of food and may be eaten by multiple predators. Food webs provide a more realistic representation of energy flow than a single food chain. Changes to one species can affect many others.

59. Producer

A producer is an organism capable of making its own organic food from simpler substances. Most producers use sunlight through photosynthesis, while some microorganisms use chemical energy through chemosynthesis. Producers form the foundation of many ecosystems because they introduce energy into biological food networks. Plants and algae are common examples.

60. Consumer

A consumer is an organism that obtains energy and nutrients by feeding on other organisms or organic matter. Consumers cannot usually manufacture all the food they need from inorganic substances. They include herbivores, carnivores, omnivores, and many decomposer-associated organisms. Consumers help transfer energy through ecosystems.

61. Decomposer

A decomposer is an organism that breaks down dead organisms and organic waste into simpler substances. Fungi and many bacteria are important decomposers. Their activity releases nutrients back into soil, water, and other parts of the environment. Decomposition is essential for nutrient cycling and ecosystem functioning.

62. Oxygen

Oxygen is a chemical element with the symbol O and atomic number 8. It is a highly reactive gas that makes up about one-fifth of Earth's atmosphere. Oxygen is essential for aerobic cellular respiration in humans and many other organisms. It also participates in combustion, oxidation reactions, and the formation of many compounds.

63. Carbon Dioxide

Carbon dioxide is a colorless gas composed of one carbon atom and two oxygen atoms. It is released through respiration, combustion, decomposition, and volcanic activity and is absorbed by plants during photosynthesis. Carbon dioxide is also a greenhouse gas that influences Earth's climate. Its concentration in the atmosphere affects the global carbon cycle.

64. Water Cycle

The water cycle describes the continuous movement of water between Earth's surface, atmosphere, oceans, soil, and living organisms. Major processes include evaporation, condensation, precipitation, infiltration, runoff, and transpiration. Solar energy drives much of this cycle. The water cycle helps distribute freshwater and regulates weather and ecosystems.

65. Evaporation

Evaporation is the process in which molecules at the surface of a liquid gain enough energy to escape into the gas phase. It can occur below the liquid's boiling point and generally becomes faster as temperature or surface area increases. Evaporation is a major part of the water cycle. It also produces cooling when higher-energy molecules leave a surface.

66. Condensation

Condensation is the process in which a gas changes into a liquid after losing enough thermal energy. In Earth's atmosphere, water vapor can condense into tiny droplets that form clouds. Condensation also occurs on cool surfaces when warm moist air contacts them. It is the reverse phase change of evaporation.

67. Precipitation

Precipitation is water that falls from the atmosphere to Earth's surface in forms such as rain, snow, sleet, or hail. It develops when atmospheric water droplets or ice crystals become large enough to fall under gravity. Precipitation is a major component of the water cycle. It replenishes rivers, lakes, groundwater, and soil moisture.

68. Atmosphere

The atmosphere is the layer of gases surrounding a planet and held by its gravitational field. Earth's atmosphere is composed mainly of nitrogen and oxygen, with smaller amounts of gases such as carbon dioxide and water vapor. It supports life, influences weather and climate, and protects the surface from some harmful radiation and space debris.

69. Climate

Climate refers to the long-term patterns and averages of atmospheric conditions in a particular region or across the planet. It includes temperature, precipitation, humidity, wind, and seasonal variations. Climate is measured over long periods rather than individual weather events. Natural processes and human activities can both influence climate.

70. Weather

Weather describes the short-term state of the atmosphere at a specific place and time. It includes conditions such as temperature, humidity, air pressure, wind, cloud cover, and precipitation. Weather can change rapidly because of atmospheric processes. Meteorologists observe and model these conditions to produce forecasts.

71. Greenhouse Effect

The greenhouse effect is a natural process in which certain gases in an atmosphere absorb and re-emit infrared radiation, warming the surface and lower atmosphere. Water vapor, carbon dioxide, methane, and other gases contribute to this effect. Without the natural greenhouse effect, Earth would be much colder. Human activities have strengthened it by increasing greenhouse gas concentrations.

72. Global Warming

Global warming refers to the long-term increase in Earth's average surface temperature, primarily associated with increasing concentrations of human-produced greenhouse gases. Burning fossil fuels, land-use changes, and other activities contribute to these emissions. Warming can influence sea levels, ecosystems, weather patterns, and the frequency or intensity of some climate-related hazards.

73. Renewable Energy

Renewable energy is energy obtained from sources that are naturally replenished on human timescales. Examples include sunlight, wind, flowing water, geothermal heat, and sustainably managed biological resources. Renewable energy can reduce dependence on finite fossil fuels. Its environmental impacts depend on the technology, location, materials, and method of deployment.

74. Solar Energy

Solar energy is energy received from the Sun in the form of electromagnetic radiation. It can be converted into electricity using photovoltaic cells or into heat using solar thermal technologies. Solar energy is renewable because the Sun continuously supplies radiation to Earth. Its availability varies with location, season, time of day, and weather.

75. Wind Energy

Wind energy is the kinetic energy associated with moving air. Wind turbines convert part of this kinetic energy into mechanical rotation and then into electrical energy. Wind is produced by differences in atmospheric pressure caused largely by uneven heating of Earth's surface. Wind power is renewable, although suitable sites and infrastructure are required.

76. Geothermal Energy

Geothermal energy is heat energy originating from Earth's interior. It can be used directly for heating or converted into electricity in regions where underground heat is accessible. Geothermal resources include hot water, steam, and naturally heated rocks. The availability of useful geothermal energy varies with geological conditions.

77. Electric Current

Electric current is the rate at which electric charge flows through a conductor or other medium. It is measured in amperes, where one ampere corresponds to one coulomb of charge passing a point each second. Current can be caused by moving electrons, ions, or other charged particles. Electrical circuits use current to transfer energy.

78. Voltage

Voltage is the difference in electric potential between two points. It represents the energy available per unit electric charge and can drive current through a circuit when a conducting path exists. Voltage is measured in volts. Batteries, generators, and power supplies can create voltage differences that enable electrical devices to operate.

79. Resistance

Electrical resistance is the opposition a material or component provides to the flow of electric current. It depends on factors such as material, length, cross-sectional area, and temperature. Resistance is measured in ohms. In many circuits, Ohm's law relates voltage, current, and resistance.

80. Magnetism

Magnetism is a physical phenomenon associated with magnetic fields and moving electric charges. Magnetic fields can exert forces on magnets, electric currents, and certain materials. Earth itself has a magnetic field that helps guide compasses. Magnetism is essential to technologies including electric motors, generators, speakers, and magnetic storage.

81. Electromagnetic Wave

An electromagnetic wave is a disturbance consisting of changing electric and magnetic fields that can transport energy through space. Unlike mechanical waves, electromagnetic waves do not require a material medium and can travel through a vacuum. Radio waves, microwaves, visible light, X-rays, and gamma rays are all electromagnetic waves with different frequencies.

82. Light

Light is electromagnetic radiation that can be detected by the human eye. Visible light occupies only a small portion of the electromagnetic spectrum, between ultraviolet and infrared radiation. Light can behave as both a wave and as discrete packets called photons. It can be reflected, refracted, absorbed, transmitted, or scattered by matter.

83. Sound

Sound is a mechanical wave produced when a source causes particles in a medium to vibrate. It can travel through solids, liquids, and gases but cannot propagate through a vacuum. Sound has properties such as frequency, wavelength, amplitude, and speed. Human hearing detects only a limited range of sound frequencies.

84. Frequency

Frequency is the number of complete cycles or repeated events that occur during one unit of time. It is measured in hertz, where one hertz means one cycle per second. Frequency is used to describe waves, vibrations, electrical signals, and periodic motion. Higher frequency means more cycles occur during the same time interval.

85. Wavelength

Wavelength is the distance between corresponding points on successive cycles of a wave, such as crest to crest or trough to trough. It is commonly represented by the Greek letter lambda and measured in metres. Wavelength is related to wave speed and frequency. Different wavelengths of electromagnetic radiation correspond to different types of radiation.

86. Quantum

A quantum is a discrete amount or packet of a physical quantity, especially energy. The concept is central to quantum mechanics, which describes physical behavior at atomic and subatomic scales. In many situations, energy can be exchanged only in specific amounts rather than continuously. The idea of quantization explains many microscopic phenomena.

87. Photon

A photon is a quantum, or individual packet, of electromagnetic radiation. It has energy related to its frequency and carries momentum despite having no rest mass. Photons make up all electromagnetic radiation, including visible light. Their interactions with matter are important in technologies such as solar cells, lasers, and medical imaging.

88. Relativity

Relativity is a theory of physics developed primarily by Albert Einstein that describes how space, time, motion, and gravity are related. Special relativity deals with objects moving at constant high speeds, while general relativity describes gravity as the curvature of spacetime. Relativity has been confirmed by many experiments and is important in modern physics and technology.

89. Black Hole

A black hole is a region of spacetime where gravity is so strong that, beyond its event horizon, nothing can escape to the outside, including light. Black holes can form when massive stars collapse or through other astrophysical processes. They can also grow by accumulating matter. Their effects can be detected through nearby matter, radiation, and gravitational waves.

90. Galaxy

A galaxy is a huge gravitationally bound system containing stars, planets, gas, dust, and dark matter. Galaxies vary greatly in size and shape and may be spiral, elliptical, or irregular. Our Solar System is located in the Milky Way galaxy. Galaxies can interact and merge over cosmic timescales.

91. Planet

A planet is a large astronomical body that orbits a star and is massive enough for gravity to make it approximately spherical. In the Solar System, planets have cleared most objects from their orbital neighborhoods. Planets can be rocky, gaseous, or icy depending on their composition and formation history. Thousands of planets have been identified around other stars.

92. Star

A star is a massive astronomical object made mostly of hot plasma and held together by gravity. Stars produce energy mainly through nuclear fusion occurring in their cores. This energy is released as radiation and supports the star against gravitational collapse for much of its lifetime. Stars differ in mass, temperature, brightness, and evolutionary stage.

93. Asteroid

An asteroid is a relatively small rocky or metallic body that orbits the Sun. Most known asteroids are found in the asteroid belt between Mars and Jupiter, although many have other orbits. Their sizes and shapes vary widely. Scientists study asteroids to learn about the early Solar System and assess potential impact hazards.

94. Comet

A comet is a small celestial body composed largely of ice, dust, and rocky material. When a comet approaches the Sun, heating can cause ices to vaporize and produce a surrounding coma and often a tail. The tail generally points away from the Sun because of solar radiation and the solar wind. Comets preserve material from the early Solar System.

95. Nebula

A nebula is a large cloud of gas and dust in space. Some nebulae are regions where new stars form, while others are produced by dying stars or stellar explosions. Radiation and gravity can shape nebulae into complex structures. Studying nebulae helps astronomers understand star formation and the chemical evolution of galaxies.

96. Universe

The universe includes all known space and time together with matter, energy, galaxies, and the physical laws that describe their interactions. It contains an enormous number of galaxies and continues to expand on large scales. Modern cosmology studies its origin, evolution, structure, and possible future. Observations indicate that the universe is billions of years old.

97. Microorganism

A microorganism is a microscopic biological organism or entity that is too small to be observed clearly without magnification. Microorganisms include many bacteria, fungi, protozoa, and microscopic algae, while viruses are generally considered biological agents rather than organisms. They live in almost every environment. Many are beneficial, while others can cause disease.

98. Bacteria

Bacteria are microscopic single-celled organisms belonging to the domain Bacteria. They have a simple cellular organization and generally lack a membrane-bound nucleus. Bacteria live in soil, water, air, and inside living organisms. They perform important roles in decomposition, nutrient cycling, food production, biotechnology, and human health.

99. Virus

A virus is an infectious biological agent consisting of genetic material enclosed within a protein coat and, in some cases, an additional membrane. Viruses cannot reproduce independently and must enter suitable host cells to make new virus particles. They infect organisms ranging from bacteria to humans. Some viruses cause disease, while others are used in research and biotechnology.

100. Antibody

An antibody is a specialized protein produced by certain immune cells in response to foreign substances called antigens. Antibodies recognize particular molecular structures and can help neutralize or mark pathogens for destruction. Different antibodies are highly specific to their targets. They are important in immunity, diagnostic tests, and some medical treatments.